

The Arctic Lens: From Revision to Drawings

Design-development guide | 19 September 2026

The central recommendation

Do not begin by making more rendered images or composing seven finished boards. Begin by coordinating one cell, one numbered settlement plan, and one section from the fjord to the mountain. These three drawings should become the common source for every subsequent plan, section, diagram and image.

The revision asks the project to move from a compelling suspended object to a credible inhabited environment. Its richness should come from differentiated spaces, legible journeys, material thresholds and seasonal experience, not from accumulating technologies.

This guide compares the complete English relazione, the revision notes, all seven accompanying sketches, and the three current colour presentation-board exports accessed on your workstation. The original files have not been changed. The seven-board thumbnail is a proposed layout, not evidence that those seven boards have already been drawn. Recommendations below are design proposals, not engineering certification or proof of regulatory compliance.

What the revision actually changes

Subject	Earlier emphasis	Revision direction	What needs to become a drawing
Arrival	A poetic ascent through rock and air	A working interchange for sea, road and proposed air access	Terminal plan, territorial section, transfer thresholds
Community	A general continuous interior	Recognisable collective, domestic and research settings	Numbered functional plan and furnished pod plans
Healthcare	Little spatial definition	Infirmiry near the arrival core and a short rescue transfer	Stretcher route through every landing, bridge and doorway
Voids	Wind, light and acoustic absence	Distinct environmental identities	Four comparable sections showing what stays open
Energy and water	Life-support described in prose	Visible relationships between collection, treatment and distribution	An architectural systems diagram with actual plant locations
Dwelling	One generic pod serving many functions	Different light, privacy and material conditions	Residential, laboratory and collective-space studies

Outdoor life	The landscape viewed from inside	Seasonal engagement with land, cliff and sea	Access, storage, changing, sheltered return and closure conditions
Presentation	Three broad boards	Seven thematic boards	A coordinated drawing set rather than seven enlarged collages

The notes report the professor's wish for more interior detail, light/material studies, and a close relationship between helipad, lift and emergency room. They do not independently establish that every proposed turbine, hydrogen tank, pool or outdoor activity was prescribed by the professor. Treat those as design options until confirmed.

Decisions to settle before final CAD work

One authoritative pod register

The project brief specifies exactly 32 cells: 8 laboratories, 12 residential pods, 6 eco-terraces, 2 technical cores and 4 open voids. Preserve this as the baseline unless you explicitly approve a replacement programme.

The new cluster sketch appears to contain 8 research cells, 8 logistics cells, 9 habitat cells, 5 eco-cultivation cells and 2 civic cells: 32 enclosed cells, with another 4 yellow void cells outside those groups. Read literally, that is 36 positions, not the stated 32. It also replaces the original programme distribution.

Use stable identifiers: L01–L08, R01–R12, E01–E06, T01–T02 and V01–V04. Write each identifier on every relevant drawing. A core symbol, furniture group or mezzanine is not another pod; an additional polyhedral envelope is.

My proposed starting approach is to incorporate civic uses, wellness and medical support as secondary spaces within the existing allocation, rather than immediately adding cells. For example, test a collective room within adjoining residential/social decks and a medical support suite near the arrival interface. This is a fit-test, not a claim that all the new uses already fit.

The population also needs a drawn test: 120 residents across 12 residential pods implies an average of 10 beds per residential pod. Draw those beds, access routes, storage and sanitary provision at scale. If the arrangement is unsatisfactory, reduce or redistribute secondary uses before adding architectural spectacle.

Continuous field or dispersed islands?

The original brief describes close-packed, face-connected cells with Ø 3,600 mm union portals. The new diagram describes four separated clusters connected by long spokes to a Civic Heart and an outer bridge ring. These are different spatial organisations.

I recommend retaining a continuous horizontal field with distinguishable functional neighbourhoods. Let shared spaces occur at the junctions between those neighbourhoods, rather than treating the Civic Heart as an extra pair of cells added after the count is complete. Keep transparent tensile bridges where a real gap or the cliff interface requires them.

If the dispersed four-cluster option is preferred, revise the written concept accordingly and measure the added bridge lengths, travel distances, exposed envelope and service routes. Do not describe long corridor-like connections as direct face-to-face continuity.

Geometry, scale and the meaning of “equatorial”

The current dwelling board uses +0.00, +8.00 and +16.00 m occupied levels and a +24.00 m upper datum. The relazione alternates between three levels and four strata. Neither should simply be relabelled to match a smaller cell: the plans and sections need reconstruction from a coordinated model.

Use the current project baseline of Level 0 for mechanical distribution, +3.5 m for the social/research deck, and +7.0 m for the upper sleeping platform. Retain 12 m as the brief’s nominal cell size and 4.2 m as its stated edge, but define exactly what that nominal diameter measures.

For a regular truncated octahedron with a 4.2 m edge, calculated dimensions are approximately 11.88 m between opposite square faces, 10.29 m between opposite hexagonal faces and 13.28 m between opposite vertices. Its full geometric surface is approximately 472.5 m², not the 282.6 m² stated in the relazione; these are whole-solid measurements before subtracting shared faces or openings. The formulas and radius/face relationships are documented by [Wolfram MathWorld](https://mathworld.wolfram.com/TruncatedOctahedron.html).

Consequently, “12 m diameter” must not silently mean all of these dimensions. Also check that +3.5 m is genuinely equatorial in the chosen orientation and local datum; the label does not make it so. A 24 m-tall section cannot represent this regular 4.2 m-edge cell.

Model one cell first, choose its orientation, cut the three floor planes, and test two or three face-connected copies. Check that the union opening provides a level, accessible walking connection: a circular Boolean cut alone does not supply a floor, threshold or usable headroom. Where a shared face is inclined, inspect its connection in section rather than assuming the plan hexagons guarantee level continuity.

Two height systems, not one

Use a site datum tied to the fjord reference level and a separate local pod datum. Give both on the territorial section and state the conversion between them.

The documents combine a settlement “22 m above the fjord”, a 45 m ascent core, a sea cable-car ascent of about 40 m, and a terminal described near the summit. These descriptions do not yet establish a consistent vertical sequence. Define whether 22 m means the lowest structural point, Level 0, or another reference. Do not assume the 45 m shaft equals the everyday passenger travel between terminal and bridge.

Draw the pier, terminal, core landings, bridge deck, pod levels and summit on one section. Retain the 45 m core as the project brief’s design parameter, but test how it can belong to the actual site geometry; do not invent surveyed elevations to force the diagram to fit.

The revised arrival sequence

For the next drawing iteration, use the following functional sequence as a proposal:

- **Road arrival:** E10 connection → mountain terminal → reception/transfer → cliff core → ETFE-enclosed horizontal bridge → +3.5 m settlement deck.
- **Marine arrival:** floating pier → protected transfer/cable-car proposal → terminal → the same core and bridge sequence.
- **Air arrival:** proposed rock-supported landing area → controlled transfer to the vertical circulation system → landing serving the settlement bridge.
- **Medical transfer:** infirmary → short, unobstructed bridge route → dedicated or shared suitably designed medical-transfer landing → air/road evacuation route.

The sketch saying that helipad, lift and hospital lie on one vertical line conflicts with the section showing the infirmary in the suspended settlement, reached by a diagonal connection to the terminal. A rock shaft cannot also be vertically coincident with a room hanging away from the cliff.

My preferred study is a rock-side transfer landing aligned with the settlement's +3.5 m bridge, minimising detours for the infirmary. Whether the express lift shares the main shaft, uses a separate shaft, or has an additional stop must be resolved in section. Draw the stretcher route, including turning spaces, rather than relying on a red arrow.

Treat “one entrance” as one normal reception point, not a demonstrated single emergency exit. An outer circulation loop only offers meaningful redundancy if routes ultimately reach suitably independent places

of safety. Fire strategy, rescue lifts, helipad operation and weather limits require specialist review; solid rock alone is not evidence of an all-weather safe landing area.

How to develop the four voids

The architectural value of the void is its unoccupied volume: sky, wind and light can enter the field.

Equipment should remain subordinate to that spatial role. Each study should show open volume, occupied edges, neighbouring thermal boundaries and maintenance access.

Void	Retain	Recommended refinement	Draw now
V01: wind and listening	Wind passage, observation, wind-harp identity	Separate any experimental turbine zone from visitor access; do not place people among rotating equipment	Section with open wind path, protected observation edge and equipment exclusion zone
V02: seasonal light	Aurora viewing and a summer-light relationship	Keep PV at selected edges or on a separate peripheral support, not across the principal sky view	Summer/winter sections and a plan of the open viewing area
V03: water and cultivation light	Daylight to planting and a collection role	Keep the well predominantly open; locate heavy storage and treatment in a defined service zone, not as decorative tanks filling the void	Collector-to-treatment-to-storage route and adjacent planted section
V04: civic oculus	Light and sky beside the common room	Study an offset small-drone landing ledge rather than a roof covering the oculus; separate people and delivery handling	Plan and section distinguishing open sky, occupied deck and landing zone

The notes place hydrogen and heat storage both in V03 and in Logistics. Select one coherent location before drawing distribution routes. I would keep hydrogen as an optional systems study rather than a fixed thesis commitment until storage location, access and safety separation are demonstrated.

Likewise, rename “power and water now solved” as “proposed resource strategy”. Show collection, conversion, storage, distribution, waste and backup as separate stages. Wind and solar symbols do not establish annual autonomy; rain and snow collection do not establish potable-water sufficiency. Reserve room for maintenance, redundancy and imported supplies without undermining the low-impact ambition.

Do not treat Moskstraumen as a whirlpool immediately beneath the project. Kartverket locates it between Lofotodden and Mosken, so it must be geographically checked against the stated site before it is used as an adjacent attraction or direct energy source ([Kartverket](#)).

Make the interiors genuinely different

Use one coherent cell envelope but several spatial interpretations. The drawings should explain light, occupation and material thickness together.

Residential pod: the protected horizon

At Level 0, draw service access and distribution rather than repeating the entire settlement plant in every pod. At +3.5 m, place the shared living area where it can receive lateral light without turning the principal circulation path into somebody's bedroom. At +7.0 m, test the actual beds and their headroom beneath the transparent apex.

Maintain a clear apex as the architectural premise. Study retractable local bed canopies or internal blackout devices as secondary elements, so that sleep control does not permanently erase the sky opening. Include storage, clothing, bathroom access and places to sit: a residential plan is more than an arrangement of bed symbols.

Deliver three plans at 1:100, a section at 1:50, one inhabited perspective and one privacy/light-control detail.

Research pod: a room that can be used

Choose one laboratory activity, for example receiving and examining marine samples, and trace its sequence from arrival to bench, storage and disposal. Draw equipment footprints and staff movement with sample-handling and service access, rather than adding the word "wet lab" to an empty hexagon.

Keep the settlement's spatial continuity as an architectural ambition, but identify where specialised laboratory operations may need controlled boundaries. Do not claim unrestricted shared air between every lab, dwelling and service room is already resolved.

Deliver a furnished working-deck plan at 1:100, a service section at 1:50, and a short sample-flow diagram. Label equipment capacities as provisional until selected.

Civic space: shared light, not an extra pod

Study the assembly space first within the 32-cell allocation. Stepped seating and a tall central volume could give it an identity without duplicating the generic pod interior.

A triple-height hall removes occupied floor area. If you pursue it, show the edges of the +3.5 m and +7.0 m layers as galleries or partial platforms, and mark this as a typological exception requiring approval rather than presenting three complete floors and an uninterrupted full-height hall simultaneously.

Distinguish the quiet communal Sanctuary from the “Celestial Sanctuary” name used for bedrooms. Otherwise one term appears to mean two different spaces. The calm room might use dark timber, an acoustic lining and a sharply framed sky aperture; stone or water should be tested as limited accents, not assumed weightless finishes.

Wellness and outdoor life: the threshold matters

Develop a modest sauna/rest/changing sequence before committing to an outdoor heated pool. Decide where wet clothing dries, where people change, how water is collected and how the space closes in bad weather.

A direct “sauna to fjord plunge” is not a short adjacent sequence if the sauna is in a settlement suspended above the water. Either move the sea-bathing sequence to the pier/base-camp interface as a separately assessed proposal, or design a different suspended wellness experience. Do not depict a direct jump from the inhabited deck.

Treat skiing, kayaking, climbing and outdoor study as possible seasonal programmes, not guaranteed site conditions. On the next map, show a sheltered return point, equipment storage, access restrictions and route closures. Differentiate observational field stations from permanent new construction.

Assessment of the three current tavole

Current board 1: Modular Territorial System

Its strengths are the clear progression from unit to assembly and the large atmospheric fjord image. Retain the desaturated landscape and the broad distinction between territorial drawing and project experience.

The site plan gives contours considerable space but does not explain arrival, the terminal or the project-to-cliff threshold. The settlement reads as a small object inside a circle. The rendering also suggests a vertically staggered aggregation that must be reconciled with the intended horizontal field.

Reuse the concept strip after correcting geometry and aggregation. Rebuild the territorial plan with verified north, a scale bar, site boundary/extent, road and marine connections, cut lines and the actual intervention points. Keep detailed arrival information for a separate board rather than overloading the landscape image.

Current board 2: System and Structure

The suspension network is the strongest element, but similar thin cyan lines make cables, pod outlines and background geometry compete. The upper functional diagram names sectors without demonstrating their internal organisation.

The lower structural elevation carries large height intervals and staggered pod positions. It must be reconstructed from the same model as the settlement plan and the new smaller cell sections. Do not retain it merely because its graphic composition is already finished.

Separate structure and programme into different boards. On the structure board, use a restrained hierarchy for primary cables, secondary restraint, cell frames and background terrain. Add one enlarged rock connection, one cell node and a sequence of assembly/disassembly. Cable sizes and anchor numbers should be labelled as preliminary where not supported by analysis.

Current board 3: Dwelling

The board has a clear plan/section/interior relationship, and the sectional image provides atmosphere. However, its three ring-like plans are mostly zoning bands with a repeated central spiral stair. They do not yet prove furnished living spaces, specific laboratory operations or the revised arrival level.

The principal section remains 24 m tall. The plans and interior should be rebuilt, not compressed vertically or relabelled. Test a stair and lift with real clearances instead of treating the spiral symbol as a complete circulation solution.

Replace the generic multifunctional pod with a small comparative set of genuinely different interiors. Reduce luminous visual effects where they conceal floors, furniture and structural edges. The perspective should explain a space already established by the plans.

Recommended seven-board set

Keep the proposed 900 × 900 mm format. The following scales are starting suggestions, not verified scale assignments; check their physical fit against measured extents and use a scale bar on every measured drawing.

Board	Main question	Dominant drawing	Supporting evidence
01: Territory and Concept	Why suspend here?	Verified site plan, provisionally 1:1500	Corrected unit-to-field sequence, compact regional locator, one landscape image

02: Arrival and Thresholds	How does a person reach the habitat?	Continuous territorial section, provisionally 1:500	Terminal plan around 1:200, medical transfer diagram, core/bridge landing around 1:100
03: Settlement and Programme	How do 120 people inhabit 32 cells?	Furnished or clearly programmed +3.5 m settlement plan, provisionally 1:200	Numbered pod register, upper/lower level diagrams, everyday/service/emergency routes
04: Suspension and Assembly	How is the field held, connected and removed?	Structural axonometric and coordinated elevation	Load-path hierarchy, anchor and node studies around 1:20, disassembly sequence
05: Domestic and Research Interiors	What is daily life like?	Residential and laboratory plans around 1:100 with sections around 1:50	Beds, sanitary provision, benches, storage, light control and one inhabited view
06: Commons, Voids and Resources	How does absence support collective life?	Civic section beside four comparable void sections	Resource-loop diagram, service-zone plan, open/closed thermal boundaries
07: Seasonal Life and Material Atmosphere	How does occupation change through the year?	A paired seasonal sectional perspective	Wellness threshold, outdoor access map, material/light detail and operating conditions

Seven boards are sufficient as a first target. If board 06 becomes overcrowded after real drawings exist, an eighth board dedicated to resource systems is more defensible than making every drawing too small. Do not expand the count just to give each rendering its own sheet.

For the next revision, prioritise boards 02, 03 and 05 as working drawings. They answer the most consequential new questions. Finish the landscape presentation and cinematic images after the geometry and occupation are coordinated.

A consistent graphic method

Keep the existing atmospheric identity, but test technical drawings on a lighter neutral field or with stronger foreground/background contrast. Use colour sparingly for programme or route distinctions and combine it with labels or line patterns.

Give each sheet one dominant drawing and two to four supporting studies. Place explanatory notes next to the drawing they explain; avoid a compulsory bottom band of small text on every board. Print a cropped test at final physical size before accepting line weights and text sizes.

Every plan should declare level, cut height where relevant, north, scale, section markers and pod identifiers. Every section should declare datums, occupied floors, usable headroom, envelope thickness and its position on the plan. Every rendering should match those drawings.

Ten useful data tables, without invented precision

The revision notes propose ten tables but say they remain unfilled. Use the following as a working register; place only the relevant short excerpt on each board and keep longer schedules in the accompanying document.

Table	Columns to create	Evidence needed before completion
Pod register	ID; primary category; secondary use; connected neighbours	Approved 32-cell plan
Area schedule	ID; Level 0 / +3.5 / +7; net usable area; exclusions	Measured closed CAD regions
Occupancy	ID; beds; daytime users; accessibility provision	Furnished fit-test; distinguish residents from simultaneous users
Arrival interfaces	Origin; transfer; destination; datum; route length	Coordinated site plan and section
Circulation	Route; everyday/service/rescue role; clear width; alternative	Measured plans and safety review
Cell geometry	Edge; orientation; envelope dimensions; portal; floor datums	Authoritative cell model
Structural elements	Element ID; role; connection; preliminary status	Structural scheme, later specialist sizing
Envelope and materials	Zone; layer/element; optical role; maintenance access	Selected build-up and manufacturer data
Resources	Source; conversion; storage; destination; backup; status	Demand/supply study, not illustrative arrows alone
Seasonal use	Space; summer mode; winter mode; closure; sheltered alternative	Site climate evidence and programme decisions

Use “to be measured”, “proposed” and “requires verification” openly. Do not copy the old 108 t mass, 38.2% wind-load reduction, 8.5 s period, or 5,289 m² heated area into the new tables as if the revised scheme demonstrated them.

What to draw first, in practical order

1. **Canonical cell sheet:** one model with named faces, dimensions, orientation, floor cuts and transparent apex. Test a pair of cells and the Ø 3,600 mm portal in section.
2. **Numbered 32-cell diagram:** preserve the baseline count and place the four voids inside that total. Locate arrivals, medical support, commons and technical interfaces before furniture.
3. **Territorial control section:** fjord datum, pier, terminal, 45 m core, bridge, settlement and proposed helipad. Resolve the meaning of the 22 m suspension height.
4. **Main +3.5 m plan:** draw real walkable surfaces, entrances, the domestic/research relationship, service interfaces and all connections.
5. **Residential fit-test:** prove the population arrangement across 12 pods, with beds, sanitary provision, storage, stairs and privacy.
6. **Research and civic studies:** make two spaces deliberately different in use, light and section.
7. **Void and material studies:** resolve thermal edges and equipment zones while preserving the voids' open character.
8. **Board composition:** insert the coordinated drawings into the seven-board structure; redraw atmospheric images only after checking them against the model.

Stop and reconsider the scheme if the geometry does not connect, the height sequence fails, or the beds do not fit. Those are useful design findings, not graphic problems to hide.

Implications for the relazione

Retain the phenomenological ambition, Saraceno reference, luminous atmosphere and ethic of reversibility. Revise the numerical and absolute claims only after the drawings establish what is actually proposed.

Replace the description of an absolutely optimal Kelvin cell with the more precise architectural argument of repeatability and spatial aggregation: Kelvin's minimum-area conjecture was superseded by the Weaire–Phelan counterexample ([Wolfram MathWorld](https://mathworld.wolfram.com/Weaire-Phelan.html)).

Distinguish structural removability from complete ecological erasure. The proposed underground terminal, tunnel excavation, anchors and marine infrastructure require an explicit account of residual intervention; sealing a tunnel is not the same as never excavating it. Keep “Zero-Trace” as an ethical objective and show the dismantling strategy, rather than guaranteeing that no physical trace will remain.

Use future/project language for proposed surveys and specialist consultation unless they have actually occurred. The current text sometimes describes marine-biologist selection, acoustic tuning and proven performance as completed work without presenting supporting evidence.

Most importantly, update the relazione after the canonical model, programme and section have been agreed. Otherwise the prose will once again describe a different building from the drawings.

The next design decision

Confirm whether the original 32-cell programme remains binding or whether the professor has approved the new four-cluster distribution. My recommendation is to preserve the 32-cell baseline for the first coordinated redraw and test how much of the revised civic, medical and environmental programme can be integrated within it.

The goal for the next revisione is not seven perfect sheets. It is a coherent settlement plan, a believable arrival section, and two or three interiors whose differences can be seen and understood.